Data Science Certification Program

Master Python, statistics, machine learning, big data and visualization tools — and turn raw data into business decisions.

LEARNING PATH · PROGRAM SYLLABUS · REAL-TIME PROJECTS & CASE STUDIES

272 Hrs	175%	12	2
Live Instructor-Led Training	Average Salary Hike	Real-Time Industry Projects	Certifications — IBM & Microsoft

01 Learning Path

1. **Cohort Orientation & Preparatory Session** — 8 Hrs

   Introduction to data tools, real-time & capstone projects, and how data is reshaping career opportunities and industrial operations. Includes fundamentals of programming and statistics for non-programmers.

2. **Python Programming — Basic to Advanced** — 40 Hrs

   Python, Anaconda, GitHub and Pandas fundamentals through to advanced data analysis with NumPy, Pandas, and data visualisation with Matplotlib and Seaborn.

3. **Statistics & Machine Learning** — 70 Hrs

   Descriptive and inferential statistics, hypothesis testing and linear algebra, followed by classification, regression, clustering and dimensionality-reduction algorithms with real case studies.

4. **Data Science Tools — Big Data & Visualization** — 86 Hrs

   SQL, MongoDB, Tableau and Power BI for analytics and dashboards, Big Data & Spark analytics with Hadoop, and time-series forecasting techniques.

5. **Data Management & Deployment** — 68 Hrs

   Excel for data analysis, MLOps fundamentals — version control, containerisation and CI for ML — and deploying models to production on AWS and Azure.

02 Program Syllabus

TERM 1 · PYTHON PROGRAMMING — 40 HOURS

Basic Python

- Programming basics & environment setup
- Strings, decisions & loop control
- Python data types, functions & modules
- File I/O, exception handling & regular expressions

Advanced Python

- Data analysis using NumPy and Pandas
- Data visualisation using Matplotlib and Seaborn
- 3 case studies on NumPy, Pandas & Matplotlib
- Coding assessment and a real-time project on completion

TOOLS & TECHNOLOGIES

Python · Anaconda · NumPy · Pandas · Seaborn

TERM 2 · STATISTICS & MACHINE LEARNING — 70 HOURS

Statistics

- Fundamentals of math & probability, population vs. sample
- Descriptive & inferential statistics, hypothesis testing
- Linear algebra and exploratory data analysis (EDA)
- Problem solving for CLT, T-test, Z-score, ANOVA & ANCOVA

Machine Learning

- Data preprocessing, logistic regression & evaluation metrics
- K-Nearest Neighbours, Decision Tree & Random Forest
- Naive Bayes, K-Means & Hierarchical Clustering, PCA, SVM
- Hyperparameter tuning across all algorithms

TOOLS & TECHNOLOGIES

Scikit-Learn · Matplotlib · Seaborn · Statsmodels

TERM 3 · BIG DATA ANALYTICS & VISUALIZATION — 86 HOURS

SQL, MongoDB & Tableau

- SQL/RDBMS, joins, grouping & subqueries
- NoSQL with MongoDB — schema design, Compass, CRUD
- Tableau dashboards, visual analytics & geocoding
- Deployment of predictive models in visualisation

Power BI, Spark & Time Series

- Power BI workflow, charts & table calculations
- Hadoop, MapReduce architecture & Spark SQL/DataFrames
- ARIMA modelling, detrending & multivariate time series
- Forecasting use cases: stock prices, sales & smartphone data

TOOLS & TECHNOLOGIES

SQL · MongoDB · Tableau · Power BI · Spark · Hadoop

TERM 4 · DATA MANAGEMENT & DEPLOYMENT

68 HOURS

Excel for Data Analysis

- Data formatting, sorting, filtering & data validation
- PivotTables, PivotCharts & advanced formulas
- Excel macros and automation
- Charting and dashboard-style reporting

MLOps & Cloud Deployment

- Introduction to MLOps and version control for ML
- Containerisation with Docker, orchestration with Kubernetes
- Continuous Integration (CI) for ML pipelines
- Model deployment & monitoring on AWS and Azure

TOOLS & TECHNOLOGIES

Excel	Docker	Kubernetes	AWS	Azure

03 Real-Time Projects & Case Studies

JPMorgan — Digital Banking Classification

12 Hrs

Forecast insurance premiums and power targeted marketing using Random Forest classification techniques.

Project Certified

Netflix — Content Recommendation Engine

17 Hrs

Build a regional viewer-categorisation recommendation engine using NLP on weekday/weekend activity data.

Project Certified

OLA — Demand Forecasting Model

18 Hrs

Reduce rider waiting time with a precise time-period demand forecasting model using latitude/longitude data.

Project Certified

Saudi Aramco — Drilling Site Analytics

14 Hrs

Use big-data analytics to identify the most cost-effective and productive drilling sites (logging while drilling).

Project Certified

IBM — Workforce Career Progression

19 Hrs

Identify performance variation across 14,000+ employees using regression and other ML techniques.

Project Certified

Swiggy — Marketing Conversion Analysis

21 Hrs

Automated keyword generation and audience targeting using text analytics and NLP-based research.

Project Certified

BMW — Used-Vehicle Price Forecasting

17 Hrs

Predict optimal resale market value from mileage, price changes and production dates.

Project Certified

Samsung — COVID-19 Time Series Forecasting

13 Hrs

Track human activity and forecast case/fatality trends from continuous mobile sensor data.

Project Certified

Jio — Telecom Churn Prediction

10 Hrs

Design a precise customer-churn model to identify the drivers of telecom customer dissatisfaction.

Project Certified

Amazon — CLV Recommendation System	18 Hrs
Surface top-performing electronics using live review data and Customer Lifetime Value analysis. *Project Certified*	

BOSCH — Predictive Maintenance	20 Hrs
Build advanced predictive models for assembly-stage fault detection from production-line data. *Project Certified*	

DataCo — Smart Inventory Monitoring	15 Hrs
An ML-powered inventory and stock-level monitoring system for intelligent supply-chain management. *Project Certified*	